UNIT II

Transportation & Assignment Problems

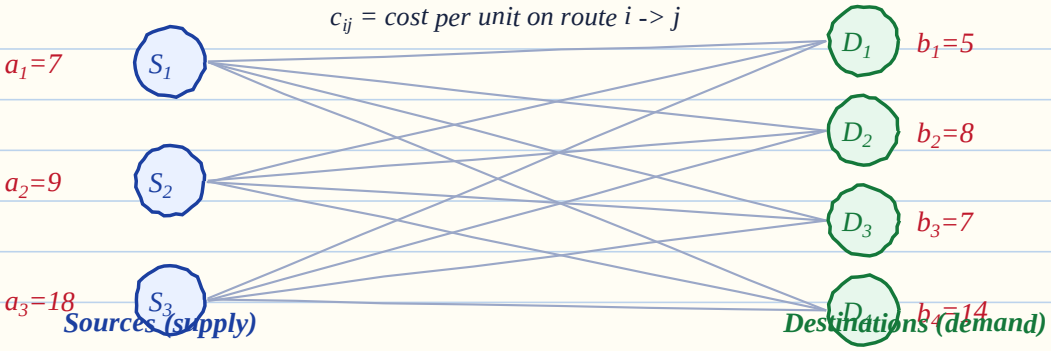
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Syllabus topics covered in this unit:

- Transportation problem: formulation, IBFS by NWC, Least Cost and VAM
- Optimality test by the MODI method
- Special cases: unbalanced and degenerate transportation problems
- Assignment problem: formulation, Hungarian method
- Traveling salesman problem

1. Transportation Problem (TP) - Formulation

The TP is a special LP in which a **single commodity** is moved from m sources (factories/warehouses) to n destinations (markets) at **minimum total transportation cost**, given the supply at each source and the demand at each destination.



Mathematical model

Minimise $Z = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij}$ (x_{ij} = units sent from source i to destination j)

Supply: $\sum_j x_{ij} = a_i$ $(i = 1..m)$ Demand: $\sum_i x_{ij} = b_j$ $(j = 1..n)$ $x_{ij} \geq 0$

Balanced TP: $\sum a_i = \sum b_j$. Number of variables = m×n, constraints = m + n.

Important facts

- A **basic feasible solution** has exactly $m + n - 1$ allocated (basic) cells; the allocations must not form a closed loop
- Always feasible & bounded when balanced. All coefficients in constraints are 0 or 1, so solutions are integers if a_i, b_j are integers
- Steps: (1) check balance (2) find **initial BFS** (NWC, LCM, VAM) (3) test optimality (**MODI**) (4) improve till optimal

Formulation example (word problem)

Two plants P1, P2 (capacity 40, 60 units) supply three markets M1, M2, M3 (demand 30, 35, 35). Unit costs (Rs): P1 -> 6, 8, 4 and P2 -> 5, 7, 9.

Let x_{ij} = units sent from plant i to market j. Min $Z = 6x_{11}+8x_{12}+4x_{13}+5x_{21}+7x_{22}+9x_{23}$

Supply: $x_{11}+x_{12}+x_{13} = 40$; $x_{21}+x_{22}+x_{23} = 60$.

Demand: $x_{11}+x_{21} = 30$; $x_{12}+x_{22} = 35$; $x_{13}+x_{23} = 35$; all $x_{ij} \geq 0$.

Total supply = 100 = total demand, hence balanced. Basic cells needed = 2 + 3 - 1 = 4.

Data used for all worked examples (cost per unit in Rs)

	D1	D2	D3	D4	Supply
S1	19	30	50	10	7
S2	70	30	40	60	9
S3	40	8	70	20	18
Demand	5	8	7	14	34 = 34

Total supply 34 = total demand 34, so the problem is **balanced**. Basic cells needed = 3 + 4 - 1 = 6.

Solution of the formulation example: VAM allocation, then MODI test.

	M1	M2	M3	Supply	u_i
P1	6, Δ=0	8, 5	4, 35	40	0
P2	5, 30	7, 30	9, Δ=6	60	-1
Demand	30	35	35		
v_j	6	8	4		

VAM cost = 540

u_1=0 gives v and u_2

Δ values shown in cells

All Δ ≥ 0 -> optimal

Optimal plan: P1->M2: 5, P1->M3: 35, P2->M1: 30, P2->M2: 30. Minimum cost = **Rs 540**.

2. Initial Basic Feasible Solution

(a) North-West Corner Rule (NWCR)

- Start at the **top-left (north-west)** cell. Allocate min(supply, demand)
- Cross out the exhausted row or column; if a row is finished move **down**, if a column is finished move **right**
- Repeat till all supplies and demands are met. Cost values are ignored, so it is fast but usually **poor**

	D_1	D_2	D_3	D_4	Supply
S_1	19 **5**	30 **2**	50	10	7
S_2	70	30 **6**	40 **3**	60	9
S_3	40	8	70 **4**	20 **14**	18
Demand	5	8	7	14	

Allocations:
(1,1)=5, (1,2)=2
(2,2)=6, (2,3)=3
(3,3)=4, (3,4)=14
Cost = 1015

Z = 5(19) + 2(30) + 6(30) + 3(40) + 4(70) + 14(20) = 95+60+180+120+280+280 = ***Rs 1015****.*
Allocated cells = 6 = m+n-1, hence non-degenerate.

(b) Least Cost Method (LCM / matrix minima)

- *Select the cell with the* ***smallest unit cost*** *in the whole table; allocate as much as possible*
- *Cross out the satisfied row/column; repeat on the remaining matrix. Ties: choose the cell that allows the larger allocation*

	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10 **7**	7
S_2	70 **2**	30	40 **7**	60	9
S_3	40 **3**	8 **8**	70	20 **7**	18
Demand	5	8	7	14	

Order of allocation:
(3,2)=8 (cost 8)
(1,4)=7 (cost 10)
(3,4)=7, (2,3)=7
(3,1)=3, (2,1)=2
Cost = 814

Z = 7(10) + 2(70) + 7(40) + 3(40) + 8(8) + 7(20) = 70+140+280+120+64+140 = ***Rs 814****.*
Better than NWCR (1015) since cost is considered.

(c) Vogel's Approximation Method (VAM / penalty method)

Gives the best starting solution; very often optimal or very close to it.

- *For every row and column find the* ***penalty = difference between the two smallest costs***
- *Select the row/column with the* ***largest penalty****; allocate max possible in its* ***least cost cell***
- *Cross out the satisfied row/column, recompute penalties for the remaining table, repeat. Ties in penalty: pick least cost cell*

Step	Row penalties S1 S2 S3	Column penalties D1 D2 D3 D4	Max	Allocate
1	9 10 12	21 22 10 10	22	x(S3,D2) = 8
2	9 20 20	21 - 10 10	21	x(S1,D1) = 5
3	40 20 50	- - 10 10	50	x(S3,D4) = 10
4	40 20 -	- - 10 50	50	x(S1,D4) = 2
5	- 20 -	- - 40 60	60	x(S2,D4) = 2
6	- 40 -	- - 40 -	40	x(S2,D3) = 7

	D_1	D_2	D_3	D_4	Supply
S_1	19 **5**	30	50	10 **2**	7
S_2	70	30	40 **7**	60 **2**	9
S_3	40	8 **8**	70	20 **10**	18
Demand	5	8	7	14	

VAM allocations:
(1,1)=5, (1,4)=2
(2,3)=7, (2,4)=2
(3,2)=8, (3,4)=10
Cost = 779

$Z = 5(19) + 2(10) + 7(40) + 2(60) + 8(8) + 10(20) = 95+20+280+120+64+200 =$ **Rs 779**.

Method	Cost (Rs)	Remark
North-West Corner	1015	Ignores costs; easiest
Least Cost	814	Uses costs; better
VAM	779	Best IBFS (optimum is 743)

3. Optimality Test - MODI (u-v) Method

MODI = Modified Distribution Method. It tests whether the current BFS is optimal using ***dual variables*** u_i *(rows) and* v_j *(columns).*

Step 1: For every ***allocated*** *cell:* $u_i + v_j = c_{ij}$. *Put* $u_1 = 0$ *(any one value) and solve for all u, v.*
Step 2: For every ***unallocated*** *cell find the opportunity cost* $\Delta_{ij} = c_{ij} - (u_i + v_j)$.
Step 3: If all $\Delta_{ij} \geq 0$ -> ***OPTIMAL*** *(*$\Delta = 0$ *for an empty cell means alternate optima).*
Step 4: Otherwise choose the most negative Δ *(entering cell), draw a closed loop through allocated cells with alternate + and -.*
Step 5: θ *= smallest allocation on the minus cells. Add* θ *at + cells, subtract at - cells. Go to Step 1.*

Applying MODI to the VAM solution (cost 779)

	D_1	D_2	D_3	D_4	Supply	u_i
S_1	19 **5**	30 Δ=32	50 Δ=60	10 **2**	7	0
S_2	70 Δ=1	30 Δ=-18	40 **7**	60 **2**	9	50
S_3	40 Δ=11	8 **8**	70 Δ=70	20 **10**	18	10
Demand	5	8	7	14		
v_j	19	-2	-10	10		

u_1=0 => v_1=19, v_4=10
u_2=50 => v_3=-10
u_3=10 => v_2=-2
$\Delta = c - (u+v)$
Δ_{22} = -18 (<0)

$\Delta_{22} = 30 - (50 - 2) = -18$ is the only negative value, so the solution is **not optimal**. Cell (S2,D2) enters.

	D_1	D_2	D_3	D_4	Supply
S_1	19 **5**	30	50	10 **2**	7
S_2	70	30 +	40 **7**	60 **2** -	9
S_3	40	8 **8** -	70	20 **10** +	18
Demand	5	8	7	14	

Closed loop:
(2,2)+ -> (2,4)-
-> (3,4)+ -> (3,2)- -> back
$\theta = min(2, 8) = 2$

Minus cells: (2,4) = 2 and (3,2) = 8, so $\theta = 2$. New allocations: $x_{22} = 2$, $x_{24} = 0$ (leaves), $x_{34} = 12$, $x_{32} = 6$.

	D_1	D_2	D_3	D_4	Supply	u_i
S_1	19 **5**	30 Δ=32	50 Δ=42	10 **2**	7	0
S_2	70 Δ=19	30 **2**	40 **7**	60 Δ=18	9	32
S_3	40 Δ=11	8 **6**	70 Δ=52	20 **12**	18	10
Demand	5	8	7	14		
v_j	19	-2	8	10		

u=(0, 32, 10)
v=(19, -2, 8, 10)
All Δ ≥ 0 -> OPTIMAL
Z = 743

Optimal solution: $x_{11}=5$, $x_{14}=2$, $x_{22}=2$, $x_{23}=7$, $x_{32}=6$, $x_{34}=12$.
$Z = 5(19)+2(10)+2(30)+7(40)+6(8)+12(20) = 95+20+60+280+48+240 =$ **Rs 743** (improved from 779).

Check: number of allocated cells must remain $m+n-1 = 6$ after every iteration. One allocated cell that became zero (x_{24}) leaves the basis.

4. Special Cases in Transportation Problems

(a) Unbalanced TP

- Supply > Demand: add a **dummy destination** with demand = excess supply; all costs in its column = 0
- Demand > Supply: add a **dummy source** with supply = shortage; all costs in its row = 0 (or shortage penalty)
- Then solve as balanced; units allotted to the dummy are actually **not shipped** (or not received)

Example: supplies 20, 30, 25 (total 75), demands 30, 25, 10 (total 65). Excess 10, so add a dummy D4 with demand 10 and zero costs.

	D1	D2	D3	D4*	Supply
S_1	8	6 **20**	10	0	20
S_2	9 **5**	7 **5**	4 **10**	0 **10**	30
S_3	3 **25**	4	2	0	25
Demand	30	25	10	10	

Optimal (VAM + MODI)
S1->D2 : 20
S2->D1:5, D2:5, D3:10
S2->dummy: 10 (unused)
S3->D1 : 25
Z = Rs 315

(b) Degenerate TP

*A TP is **degenerate** if the number of allocated cells is **less than m + n - 1** (it happens when a row and a column are exhausted at the same time). Remedy: put a very small quantity ε (≈ 0) in an **unallocated independent cell** (one that does not form a closed loop with the other allocations) and treat it as basic. At the end put ε = 0.*

	D_1	D_2	D_3	Supply
S_1	4 **10**	6 **ε**	8	10
S_2	5	3 **20**	7	20
S_3	9	2 **5**	5 **25**	30
Demand	10	25	25	

NWC: x_{11}=10 exhausts row 1 and column 1 together, only 4 cells < 3+3-1 = 5
add ε at (1,2)
then apply MODI as usual

Treat ε as a basic cell and apply MODI. Take $u_1 = 0$:

	D_1	D_2	D_3	Supply	u_i
S_1	4 **10**	6 **0**	8 Δ=-1	10	0
S_2	5 Δ=4	3 **20**	7 Δ=1	20	-3
S_3	9 Δ=9	2 **5**	5 **25**	30	-4
Demand	10	25	25		
v_j	4	6	9		

u=(0,-3,-4) v=(4,6,9)
Δ_{13} = 8-(0+9) = -1
ε cell leaves (θ = ε)
ε moves to (1,3)

After the exchange (θ = ε ≈ 0, so cost does not change) all Δ ≥ 0. Optimal cost = 10(4) + 20(3) + 5(2) + 25(5) = 40+60+10+125 = ***Rs 235****, with ε = 0.*

(c) Other variations

- *Maximisation TP (profit given): convert by subtracting every profit from the largest profit, or multiply by -1, then minimise*
- *Prohibited route: assign a very large cost M to that cell so that it never gets an allocation*
- *Multiple optimal solutions: an unallocated cell with $\Delta = 0$ in the optimal table*
- *Transhipment (intermediate points) and capacity-restricted routes are treated as extensions of the TP*

Rules for the closed loop (MODI)

- *Loop starts and ends at the entering (empty) cell; every other corner is an **allocated** cell*
- *Moves are only **horizontal or vertical**; each turn is at an allocated cell; alternate signs + - + - ...*
- *The loop may jump over (cross) other cells, but cannot turn there. For a given non-degenerate BFS the loop is **unique***

Maximisation TP - example idea

*Profit per unit: S1 -> (12, 18, 6), S2 -> (14, 3, 10). Convert: largest profit = 18, so new costs = 18 - profit = (6, 0, 12) and (4, 15, 8). Solve as a normal minimisation; the final Z is computed from the **original profits**.*

TP versus AP - short comparison

Point	Transportation	Assignment
Supply / demand	any positive numbers	exactly 1 each
Matrix	$m \times n$ (rectangular)	$n \times n$ (square)
x_{ij}	any non-negative	0 or 1 only
Basic cells	$m + n - 1$	n assignments
Method	NWC/LCM/VAM + MODI	Hungarian method

Applications

- *TP: distribution of goods from factories to depots, allocation of fuel/raw material, scheduling of production among plants*
- *AP: assigning workers to machines, salesmen to territories, cabs to customers, teachers to classes, contracts to bidders*

5. Assignment Problem (AP)

*The AP is a special TP in which n jobs are assigned to n workers (machines) on a **one-to-one basis** so that total cost (time) is minimum (or total profit is maximum). Each supply = each demand = **1**.*

Min $Z = \sum_i \sum_j c_{ij}\, x_{ij}$ *where* $x_{ij} = 1$ *if worker i is assigned job j, else 0*

$\sum_j x_{ij} = 1$ *(each worker one job)* $\sum_i x_{ij} = 1$ *(each job one worker)*

- Square cost matrix ($n \times n$). Number of assignments = n. TP methods work, but the **Hungarian method** is much quicker (degenerate TP otherwise)

Hungarian method - algorithm (Kuhn / Flood)

- 1. If the matrix is not square, add a dummy row/column of zeros. For maximisation, subtract all entries from the largest entry
- 2. **Row reduction:** subtract the smallest element of each row from all elements of that row
- 3. **Column reduction:** subtract the smallest element of each column from that column
- 4. Cover **all zeros** with the **minimum number of horizontal/vertical lines**
- 5. If lines = n -> optimal assignment exists: go to step 7. If lines < n go to step 6
- 6. Find the smallest **uncovered** element k. Subtract k from all uncovered elements, add k to elements at line intersections, leave others. Return to step 4
- 7. Assign: start with a row/column having a **single zero**, tick it, cross out other zeros of that row and column; repeat

Example: assign 4 jobs to 4 men, minimise cost (Rs)

Given matrix

	J1	J2	J3	J4
M1	15	15	23	25
M2	17	39	18	16
M3	17	29	24	6
M4	28	31	15	14

Step 2 - row reduction (row minima 15, 16, 6, 14 subtracted)

	J1	J2	J3	J4
M1	0	0	8	10
M2	1	23	2	0
M3	11	23	18	0
M4	14	17	1	0

Step 3 - column reduction (column minima 0, 0, 1, 0). Now cover all zeros: row M1, column J3 and column J4 -> only 3 lines < 4, so the assignment is not yet possible.

	J1	J2	J3	J4
M1	0	0	7	10
M2	1	23	1	0
M3	11	23	17	0
M4	14	17	0	0

Step 6 Smallest uncovered element = 1. Subtract 1 from uncovered cells (rows M2-M4, columns J1-J2), add 1 where lines cross (M1,J3) and (M1,J4).

	J1	J2	J3	J4
M1	0	0	8	11
M2	0	22	1	0
M3	10	22	17	0
M4	13	16	0	0

Now four lines are needed -> optimal. Assign: ***M3->J4*** *(only zero in row M3),* ***M4->J3****,* ***M2->J1****,* ***M1->J2****.*

Optimal assignment: *M1->J2 (15), M2->J1 (17), M3->J4 (6), M4->J3 (15).*
Minimum cost = 15 + 17 + 6 + 15 = Rs 53

Variations of the assignment problem

- *Maximisation: subtract each element from the maximum element of the matrix, then minimise*
- *Unbalanced (m ≠ n): add dummy rows/columns with zero cost. Job assigned to a dummy is left undone*
- *Restricted/prohibited assignment: put a very large cost M in that cell*
- *Multiple optimal solutions: when, in the final matrix, there are alternative sets of independent zeros*
- *Crew assignment / sales territories: typical practical uses; the Hungarian method works for any square cost matrix*

Maximisation example (profit matrix, Rs)

Profit of assigning worker W_i to job J_j:

	J1	J2	J3	J4
W1	16	10	14	11
W2	14	11	15	15
W3	15	15	13	12
W4	13	12	14	15

Largest element = 16. Subtract every element from 16 to get the loss (opportunity) matrix, then do row reduction (column minima are all 0):

	J1	J2	J3	J4
W1	0	6	2	5
W2	1	4	0	0
W3	0	0	2	3
W4	2	3	1	0

Four independent zeros exist: W1->J1, W2->J3, W3->J2, W4->J4.
Maximum profit = 16 + 15 + 15 + 15 = Rs 61 (read from the ORIGINAL profit matrix).

Unbalanced assignment (3 men, 4 jobs)

Add a **dummy man** (row of zeros) to make the matrix 4 × 4. After row reduction, three lines cover all zeros (< 4), smallest uncovered element = 2; adjusting gives:

	J1	J2	J3	J4
M1	0	0	7	1
M2	0	3	0	3
M3	1	1	0	0
Dummy	2	0	0	2

Assign M1->J1 (10), M2->J3 (7), M3->J4 (11), dummy->J2 (J2 stays undone). Cost = 10 + 7 + 11 = **Rs 28** for original matrix [10 12 19 11 / 5 10 7 8 / 12 14 13 11].

Restricted assignment

If worker i cannot do job i (diagonal prohibited), put M (very large) on the diagonal and solve normally. Example matrix [M 6 9 7 / 8 M 5 4 / 7 6 M 8 / 9 4 7 M] gives W1->J4, W2->J3, W3->J1, W4->J2 with total cost 7+5+7+4 = **23**.

6. Traveling Salesman Problem (TSP)

A salesman must **visit each of n cities exactly once** and return to the starting city so that the total distance (cost/time) is minimum. Its model is like an assignment problem with the extra condition that the assignment forms **one single tour**.

- Cost matrix c_{ij} (distance city i to city j); diagonal $c_{ii} = \infty$ (M), since a city cannot be followed by itself
- (n - 1)! possible tours for an asymmetric problem - too many to list when n is large

Solution procedure (assignment + sub-tour elimination)

- Solve as an AP with the Hungarian method ignoring the tour condition
- If the solution gives **one cycle through all cities** -> optimal tour
- If it splits into **sub-tours**, use branch-and-bound: forbid one arc of the sub-tour (cost M)

and re-solve, or patch the sub-tours using the least-cost link

Example: 5 cities A, B, C, D, E (distances)

	A	B	C	D	E
A	∞	3	6	2	3
B	3	∞	5	2	3
C	6	5	∞	6	4
D	2	2	6	∞	6
E	3	3	4	6	∞

Hungarian method on this matrix gives the assignment A->B, B->D, D->A and C->E, E->C with cost 3+2+2+4+4 = 15. This is ***not a tour****: it has two sub-tours A-B-D-A and C-E-C. So 15 is only a lower bound.*

Break the sub-tours by branching (forbid an arc such as D->A, re-solve, or link the cycles through the cheapest exchange). The best complete tour is:

A -> D -> B -> C -> E -> A *with distance 2 + 2 + 5 + 4 + 3 =* ***16 units****.*
(Verified by comparing all (5-1)! = 24 tours; no tour is shorter than 16.)

Feature	Assignment	TSP
Solution	any permutation	one single cycle (tour)
Extra condition	none	no sub-tours
Method	Hungarian	Hungarian + branch & bound

7. Transshipment Problem & Method Comparison

In a ***transshipment problem*** *goods may move from a source to a destination* ***through intermediate points*** *(other sources or destinations). Total cost = sum of costs along the chosen route.*

- *Treat every source and destination as both a possible origin and possible destination*
- *Buffer stock B = total supply (or total demand) is added to every supply and demand figure, so each point can hold flow in transit*
- *Cost from a point to itself = 0. Solve the enlarged balanced TP by VAM + MODI; flow on c_{ii} cells is just a balancing quantity*

Merits and demerits of the IBFS methods

Method	Merits	Demerits
NWCR	Very simple, no thinking	Ignores cost; far from optimum
LCM	Uses cost; quick; good start	Local choice, may miss better plan
VAM	Best start; fewest MODI steps	More calculation (penalties)

Common mistakes to avoid (exam tips)

- Forgetting to balance the problem first (dummy row/column with ZERO cost)
- Having fewer than $m + n - 1$ allocations and not adding ε; or adding ε to a cell that forms a loop
- Taking u, v from empty cells - they must be computed only from allocated cells
- In MODI, forgetting that θ is taken from the minus cells only, and the leaving cell has the smaller allocation
- In Hungarian method: covering zeros with more lines than the minimum; not adding k at intersections
- Maximisation AP: reading the final answer from the transformed (loss) matrix instead of the original profits
- TSP: leaving sub-tours in the answer - a valid tour visits each city exactly once

Worked mini-check: is a given plan optimal?

Given an allocation with $m + n - 1$ cells: (1) write $u_i + v_j = c_{ij}$ for each allocated cell, (2) compute Δ for empty cells, (3) if no $\Delta < 0$ the plan is optimal. If one $\Delta = 0$ there is an alternate optimum with the same cost, obtained by bringing that cell in through its loop.

8. Quick Revision & Important Questions

1. Give the mathematical form of a TP; explain NWC, LCM and VAM with an example.
2. MODI method steps; how to deal with unbalanced and degenerate TP.
3. Formulate the assignment problem; Hungarian method with maximisation variation.
4. Explain the TSP and its relation to the assignment problem.

Remember: basic cells = $m + n - 1$; $\Delta_{ij} = c_{ij} - u_i - v_j$; optimal if all $\Delta \geq 0$; VAM penalty = difference of two lowest costs; Hungarian: lines = n => stop.

Question bank (university pattern)

2-marks: define balanced TP; why dummy is added; what is degeneracy in TP; why Hungarian method needs a square matrix; define sub-tour.

5-marks: VAM with an example; steps of MODI; assignment of maximisation type; unbalanced assignment problem.

10-marks: solve a 3×4 or 4×4 TP with VAM + MODI; solve a 4×4 assignment problem by the Hungarian method; TSP as an assignment problem.